

Introduction to PowerBI & Power Query

Hands-On Tutorial: Build a Report from Excel

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Project 02 – First Hands-On

M.Sc. in Electrical and Computer Engineering – ISE – Universidade do Algarve

Based on: Microsoft Learn “Create a report from an Excel workbook in Power BI”

Business Scenario

Your manager needs an executive summary that answers:

- Which month and year had the highest profit?
- Which country/region performs best?
- Which product and segment should receive more investment?

1 Learning Outcomes

By the end of this tutorial, you will be able to:

- Import Excel data into Power BI Desktop.
- Apply core data preparation steps in Power Query for cleaner analysis.
- Create visuals and a slicer for an executive dashboard.
- Answer management questions with evidence from report visuals.
- Publish and share a complete report in the Power BI Service.

2 Prerequisites

- Power BI Desktop installed.
- Basic familiarity with Excel and Power BI interface.
- Power BI account (only required for publishing).
- Sample file: `Financial Sample.xlsx`.

Direct download link: <https://download.microsoft.com/download/1/4/E/14EDED28-6C58-4055-A65C-23B4DA81C4DE/Financial%20Sample.xlsx>

3 Lab Workflow Overview

1. Get data from Excel.
2. Prepare data in Power Query.
3. (Optional) Create DAX measure and Calendar table.
4. Build report visuals.
5. Add slicer and apply formatting.
6. Save and publish.

4 Step 1 – Import Data

1. Open **Power BI Desktop**.
2. Select **Excel workbook** from the Home page.
3. Browse to `Financial Sample.xlsx` and click **Open**.
4. In Navigator, check **financials**, then click **Transform Data**.

5 Step 2 – Prepare Data (Power Query)

1. Change `Units Sold` data type to **Whole Number**:
 - Transform → Data Type → Whole Number → Replace current.
2. Format `Segment` as uppercase:
 - Transform → Format → UPPERCASE.
3. Rename `Month Name` to `Month`.

4. Filter out Montana from the Product column.
5. Click **Close & Apply**.

Checkpoint

In Query Settings, verify all transformations are listed in *Applied Steps*.

6 Step 3 – Optional DAX Enhancements

3.1 Create a Measure

1. Home → **New measure**
2. Enter:

```
Total Units Sold = SUM(financials[Units Sold])
```

3.2 Create a Calendar Table

1. Switch to Table view.
2. Home → **New table**
3. Enter:

```
Calendar = CALENDAR(DATE(2013,01,01),DATE(2014,12,31))
```

If your locale requires semicolons:

```
Calendar = CALENDAR(DATE(2013;01;01);DATE(2014;12;31))
```

3.3 Create Relationship

1. Go to Model view.
2. Drag Calendar[Date] to financials[Date].
3. Save the auto-detected relationship.

7 Step 4 – Build the Executive Report

Visual 1: Report Title

1. Insert → Text box.
2. Type: **Executive Summary – Finance Report**.
3. Set font to bold, size 20.

Visual 2: Profit by Month and Year (Line Chart)

1. Drag Profit to canvas.
2. Add Date to the same visual.
3. In X-axis, choose **Date Hierarchy**.
4. Keep only **Year** and **Month** (remove Quarter and Day).
5. Change visual type to **Line chart**.

Visual 3: Profit by Country/Region (Map)

1. Drag Country/Region to canvas.
2. Convert to **Map** if needed.
3. Add Profit to map size/value.
4. Confirm bubble map view (not filled map) if your class expects bubbles.

Visual 4: Sales by Product and Segment

1. On a blank area, select Sales, Product, Segment.
2. Keep/use clustered column chart.
3. Resize and position below Visual 2 and Visual 3.

8 Step 5 – Add Slicer

1. Drag Calendar[Date] to blank canvas area.
2. Change visual to **Slicer**.
3. Keep only **Year** and **Month**.
4. Expand year nodes and resize for readability.

9 Questions to Answer with the Report

After building the report, answer the following questions in short text (1–3 sentences each), and cite the visual(s) used as evidence:

1. Which month and year has the highest profit? What is the approximate value?
2. Which country/region contributes the most profit? Which contributes the least?
3. Which product has the highest sales overall?

4. Which segment (Government, Small Business, etc.) appears strongest across products?
5. For year 2013 only (using the slicer), what changes compared with the full-period view?
6. For year 2014 only (using the slicer), which insight is most different from 2013?
7. Based on your visuals, where should the company invest next quarter, and why?

Validation Questions

Use your visuals to answer:

- Highest profit month/year: **December 2014**
- Strongest region: **Europe (notably France and Germany)**
- Priority product/segments: **Paseo, Small Business, Government**

10 Step 6 – Optional Formatting

- View → Theme → **Executive**.
- Increase visual title sizes to improve readability.
- Enable subtle shadow effects on key visuals.
- Add a rectangle shape behind the page title for stronger header contrast.

11 Step 7 – Save and Publish

1. Save your file (Ctrl + S).
2. Home → **Publish**.
3. Choose destination (for example, **My workspace**).
4. Open in Power BI Service and use **Share**.

12 Troubleshooting Notes

- If DAX syntax fails, type formulas manually to avoid hidden characters.
- If your locale is non-English, replace commas with semicolons in DAX.
- If Date Hierarchy is missing, remove and re-add Date, then select Date Hierarchy.

13 Submission Checklist (for students)

- Data imported and transformed correctly.
- All required visuals created and arranged.
- Year/Month slicer works as expected.
- Business questions answered with evidence.
- Report saved (and optionally published/shared).

14 Key Takeaways

- Data quality and correct data types directly impact visual clarity and decision quality.
- A small set of focused visuals can answer high-value business questions quickly.
- Slicers are essential for interactive analysis and period-over-period comparisons.
- Effective reporting combines technical execution (model, visuals, publish) with clear business interpretation.

End of tutorial